



A systematic review of the pollution and health impacts of low emission zones

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ABSTRACT

Air pollution is one of the most significant issues that urban areas face, with 7 million mortalities worldwide each year as a result (UN, 2023). Different policies have been introduced to mitigate the emissions of harmful pollutants through a variety of means, including Low Emission Zones (LEZs), which have been introduced in cities to reduce the number of journeys made by highly polluting vehicles. This study uses two robust regression models to answer two research questions related to the key metrics of pollutant concentrations and long-term health benefits following LEZ implementations. The two pollutants, Nitrogen Dioxide (NO₂) and Fine Particulate Matter (PM_{2.5}), were examined with the “openair” software in R, which is used to analyse trends in air pollution. The impact that different policies have on several health metrics were also examined through data collated from literature reviews. The results obtained after analysing London’s LEZ (2008) and ULEZ (2019) were then used to evaluate the viability of Manchester’s Clean Air Plan. The results show that there is statistically significant evidence of LEZs curbing pollution levels, which in turn contributes to long-term health benefits. It is concluded that the current trajectory of Manchester’s air pollution policy, which falls short of having an LEZ, is likely not impactful enough on pollutant concentrations and health, especially in relation to PM_{2.5} levels. Future research at planned time intervals is suggested for modelling the extent of the impact of Manchester’s Clean Air Plan. However, results demonstrate the importance of curbing harmful pollutants, which should be taken into consideration by policy makers in other cities as well.

1. Introduction

Increasing vehicle emissions are the single largest source of both Nitrogen Dioxide (NO₂) and Particulate Matter (PM_{2.5}) (Pidgeon, 2020). Dangerous pollutant concentrations have a consequential impact on human health, including excess mortality; causing wide reaching social and economic impacts on society (UN, 2023). Transport-led solutions, including the implementation of Low Emissions Zones (LEZs) and similar policies, have been introduced following local, national, and international directives to improve air quality. Air pollution is a concern that affects the UK, where this study was based, with dense urbanisation contributing to many cities having unsafe air pollution levels. New policies, both in the UK and other regions, are often met with polarising support and opposition. The benefits and drawbacks of LEZs are complex and go beyond a simple cost-benefit analysis; this paper uses a vast body of existing literature and modelling in two cities (London and Manchester) to evaluate the viability of these schemes.

Nitrogen Dioxide (NO₂) and Fine Particulate Matter (PM_{2.5}) are the

two pollutants that are being assessed in this study. Due to their impact on human health, both pollutants are critical to the evaluation of LEZs. Nitrogen Dioxide is a more local pollutant than Particulate Matter, and is introduced to the environment largely from vehicle traffic (PHE, 2018). The biological impacts of NO₂ are highly damaging, with severe respiratory damage increasing the risk of diseases such as asthma and lung cancer. Fine Particulate Matter (PM_{2.5}) is produced by combustion, for example from vehicle emissions, and is an urban pollutant that often disperses over a large area (PHE, 2018). PM_{2.5} (fine particles with a diameter of 2.5µm or less) is analysed instead of PM₁₀ (particles with a diameter of 10µm or less), due to its danger to human health; PM_{2.5} particles have the capability to penetrate deep into the human respiratory system. There are several associations of over-exposure to PM_{2.5} to harmful diseases, including different types of respiratory and cardiovascular diseases. PM_{2.5} is considered to be the largest contributor to the problem of urban air pollution (Szönyi et al., 2007). The monitoring of PM₁₀ data, as well as other pollutants such as ozone (O₃), are also critical to mitigating urban air pollution; although for this study they are

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discounted from this scope to streamline the analysis.

Two main research questions are included in this study to assess the effectiveness of Low Emission Zones: RQ1) Does the implementation of Low Emission Zones in urban areas significantly reduce air pollution? And RQ2) Does the implementation of Low Emission Zones in urban areas significantly improve population health?

To answer the first research question, directional trends were assessed, as well as comparing overall emissions to the regulated standards. The UK Air Quality Regulations set in 2010 require that pollutant levels do not exceed an annual average of 40 $\mu\text{g}/\text{m}^3$ for NO_2 and 20 $\mu\text{g}/\text{m}^3$ for $\text{PM}_{2.5}$ (gov.uk, 2023a). The latest UK Environmental Targets further reduced the $\text{PM}_{2.5}$ guidelines to 10 $\mu\text{g}/\text{m}^3$ (gov.uk, 2023b); highlighting the urgency to reduce the harmful impacts of this pollutant. The most recent World Health Organization Air Quality Guidelines recommend that pollutant levels do not exceed an annual average of 40 $\mu\text{g}/\text{m}^3$ for NO_2 and 5 $\mu\text{g}/\text{m}^3$ for NO_2 (WHO, 2021). The WHO recommendations for NO_2 are consistent with the UK guidelines, although the $\text{PM}_{2.5}$ limits (5 $\mu\text{g}/\text{m}^3$) are even lower than the updated UK targets, and a fifth of the latest EU limits of 25 $\mu\text{g}/\text{m}^3$. It is worth noting that the WHO limits are not legally binding, whereas the UK and EU limits are.

LEZs typically involve restrictions on cars depending on their emission class, with charges and fines applied for driving non-compliant vehicles inside the zone boundary. The emissions standards vary for different LEZs. For example, to be exempt from the London Ultra Low Emission Zone (ULEZ) charge vehicles need to be Euro 4 or higher (NO_x) for petrol vehicles and Euro 6 (NO_x and PM) for diesel vehicles (TfL, 2024). The main difference between Euro 4 and Euro 6 emissions standards is the level of permitted pollutants from vehicles. Euro 6 standards are stricter, with significantly lower limits on nitrogen oxides (NO_x) and particulate matter (PM) emissions compared to Euro 4. Specifically, Euro 6 requires a reduction in NO_x emissions for diesel cars by more than 50 % and tighter control of PM levels for both petrol and diesel vehicles, aiming to improve air quality and reduce environmental impact. Other cities may require higher standards, such as the stricter 'Umweltzonen' (environmental zones) requirements across German cities, of which some (e.g. Munich, Darmstadt, Stuttgart) have a complete diesel ban in their low emission zone areas. Debate over whether LEZs should be implemented and what form they should take is extensive and often attracts significant political attention, leading to the widespread variations in the legislation of emissions requirements. The Volkswagen emissions scandal of 2015, which exposed the car manufacturers for rigging newly manufactured diesel vehicles to meet US emissions standards, raised questions over the consistency of the exemption requirements for LEZs (McGee, 2017). There is also debate over restrictions on smaller and larger vehicles (e.g. motorcycles, commercial vehicles), and the standards for PM emissions from abrasion emissions (see Section 5). Most LEZs are applied in the urban centres of city regions, often with ring roads used as boundaries. Critics have argued that there are undesirable spillover effects of LEZs, with background pollutant concentrations outside the boundaries of LEZs potentially increasing as drivers avoid charges and fines.

Political discourse surrounding LEZs is highly relevant, and often evolves around whether the adverse effects justify the means. Many argue that LEZs fail to be financially viable, with residents left between paying daily charges or investing in a costly vehicle upgrade. Debate is often linked with the impact of pedestrianisation and the introduction of 'Low Traffic Neighborhoods', which are urban areas where measures are taken to reduce the volume and speed of motor vehicle traffic on residential streets, with critics asserting that limiting vehicle access in regions restricts trade and opportunities. Proponents for LEZs argue that the altering of behaviors culminated following new legislation, for example modal shifts towards sustainable transportation, justify the introduction of LEZs. The effectiveness of LEZs on reducing pollutant levels in urban areas, and the rate at which they improve public healthcare, is unmistakably critical to assessing their efficacy and whether society benefits from their introduction. Although a fuller

discussion of the political discourse surrounding LEZs is beyond the scope of this paper, notable papers study the environmental justice aspects of LEZs in Antwerp (De Vrij and Vanoutrive, 2022), Brussels and London (Verbeek and Hincks, 2022), Gothenburg (Börjesson and Kristofferson, 2015), and Grenoble (Charleux, 2014).

This study focuses on the cities of London and Manchester. With the populations of Greater London and Greater Manchester around 9 million and 3 million respectively (ONS, 2024), they are highly relevant to urban air pollution strategy in the UK. The alternative routes the two regions have approached tackling urban air pollution has attracted significant political attention, with the mayors for both cities (Sadiq Khan in London and Andy Burnham in Manchester) receiving scrutiny and praise from different groups regarding their strategies. Both regions have had problems air quality issues over the years. In London, it was estimated in 2015 that nearly 9,500 people died prematurely each year because of urban air pollution (Walton et al., 2015), although evidence from several studies suggest that there has been a significant decrease in pollutant levels across the capital since then. In Manchester, it is estimated that there are 1,200 premature deaths every year due to air pollution (cleanairgm.com, 2024).

In London, the first LEZ was introduced in February 2008, which charges a fee to heavy vehicles (e.g. lorries over 3.5 tonnes, coaches) that do not meet emissions standards. The London LEZ still exists and covers most of Greater London, and currently requires Euro 6 emission standards for lorries, buses and coaches. In 2017, the toxicity charge, or T-charge, was introduced in central London (in the same area and operating at the same hours as the congestion charge), charging older cars that did not meet Euro 4 standards an extra £10 per day. This T-zone charge was on top of the congestion charge, which was introduced in 2003 and applies to most vehicles to drive in central London (within the London Inner Ring Road) during selected hours. Active 24-hours a day, the Ultra Low Emission Zone (ULEZ) was introduced in April 2019 and replaced the short-existing T-charge. Applying initially in the same area as the central London Congestion Charge Zone, the ULEZ charges £12.50 a day for each non-exempt vehicles that uses the zone (Euro 6 for diesel vehicles, Euro 4 for petrol vehicles, Euro 3 for motorcycles). The ULEZ has since had two expansions and now covers most of Greater London, in 2021 (to cover most of inner London) and 2023 (to cover almost all areas within the M25).

Unlike in London, Greater Manchester has never had a Low Emission Zone or a similar scheme. To meet the UK legal pollutant guidelines by 2024, a Category-C Clean Air Zone (CAZ) was due to be introduced in Manchester in May 2022. As a Category-C CAZ, it would have introduced charges for non-compliant coaches, taxis, HGVs and vans but not to private passenger cars. However, following the Covid-19 pandemic's significant impact on vehicle supply chain issues and businesses, Greater Manchester's elected local authorities shifted towards being in favour of using 'investment-led' strategies to meet the guidelines. Just three months before the CAZ was due to be introduced, the policy was discarded. Part of the argument presented by Manchester's authorities was that the CAZ would not have met legal directions to cut emissions in time due to the pandemic (Brown, 2023), stating that charging clean air zones are a "pre-pandemic solution for a post-pandemic world." Significant investment into the original CAZ had occurred before its suspension, including spending over £18 million on installing Automatic Number Plate Registration (ANPR) cameras and over £2 million on signage (cleanairgm.com, 2024).

Instead, Manchester has introduced an ongoing 'Clean Air Plan'. This includes schemes such as targeted public vehicle upgrade funding (e.g. the 'Clean Taxi Fund'), delivering a new zero-emission 'Bee Network' bus network, and investment in local traffic measures. The current plan has received significantly less opposition than the original CAZ, with the removal of charges for non-compliant vehicles popular with residents. It is worth emphasising that the original Category-C CAZ would not have included charges for private cars and motorcycles, aligning the plan more with London's 2008 LEZ instead of the recently introduced ULEZ.

The relevant literature is discussed below, followed by an explanation of the materials and methods involved in studying the two research questions. Section 4 Results introduces this paper's findings, which are discussed in Section 5 Discussion, in which the limitations of this paper and recommendations for future policy are also explored. Section 6 Conclusions summarises the contributions of this paper and provides suggestions for follow-up research.

1.1. Brief literature review

This study relies on the existing literature for data. There is an extensive body of literature assessing the viability of Low Emission Zones and other schemes to tackle urban air pollution. This section focuses on literature that is related to answering the two research questions, although several other factors (e.g. the impact of LEZs on modal shift) have also been examined. The assessment of LEZs is convoluted and does not necessarily have a clear-cut process for creating a cost-benefit analysis. This is illustrated in this section by the several approaches to methodology different reports have taken.

Existing literature highlighted that regression analysis is usually used to determine whether LEZs and similar schemes are effective in curbing pollution levels and improving long-term public health. Difference-in-differences approaches were the most common, and involve using a treatment group and control group to estimate the causal effect of the treatment (LEZs) over time. The use of the openair package (Carslaw and Ropkins, 2012) and Theil-Sen trends to analyse trends in air pollution in the Ellison et al. (2013) report was incorporated into answering RQ1. As well as lots of research into London's zonal charging policies existing, there is also an extensive body of work into German LEZs. Gehrsitz (2017) highlights that the use of German LEZs to assess policy measures is effective due to the staggered introductions of LEZs across different cities, as well as the wide range of LEZ adopting and LEZ non-adopting areas. However, this study uses London and Manchester as its focus cities, particularly due to the lack of existing research into Manchester's air pollution and Clean Air Zone policy. London is also highly suitable for assessing the adoption of LEZ policies, as cited in the Ma et al. (2021) paper that the ULEZ is the "the world's most stringent emission zone".

Table 1 presents the key research outcomes from modelling impact of LEZs on different pollutants. There are many inconsistencies with the rates of pollution reduction across the different reports. Table 1 also shows that (particularly for German reports) PM_{10} is often used to analyse trends. Although this study opts to analyse trends in $PM_{2.5}$, this pollutant is still very harmful and is considered when examining existing literature. Beshir and Fichera (2022), found that the ULEZ caused significant reductions in daily PM_{10} levels (18 % reduction compared to the baseline mean) and daily NO_2 levels (12 % reduction compared to the baseline mean). Wolff (2014) examines the impact of German LEZs on reducing particulate emissions and vehicle fleet turnover. It is found that German LEZs decrease PM_{10} levels by 9 %, and rejects the hypotheses that dirty vehicles contribute to higher pollution by taking longer routes outside zone boundaries due to restrictions. Despite only finding what is described as marginal impacts of LEZs on pollution, ranging from -2.5 % to -8.5 %, Bernado et al. (2021) also concludes that there is no evidence of LEZs exporting polluting cars from city centres to unrestricted areas. This study is a systematic review using econometric analysis techniques to investigate different LEZs across Europe, and discusses that LEZs are the most effective when they are stricter (e.g. they are applied to a wide area of the city and are stringent in the type of unrestricted vehicles). Wolff (2014) tends to agree with this conclusion, stating that alternative approaches to LEZs (e.g. building ring roads) lead to negligible decreases in air pollution.

Not all existing literature found statistically significant reductions in pollutant concentrations following the introduction of LEZs. Utilising a regression discontinuity design model, Ma et al. (2021) examined the short-term impact of the 2019 ULEZ in London. In the context of a longer-term downward trend in London's air pollution, this study only

Table 1

Summary of key results from literature review relating to RQ1 (pollutant concentrations).

Title of Paper	Author	Year	Key Results (relating to RQ1)
"And Breathe Normally": The Low Emission Zone impacts on health and well-being in England.	Beshir and Fichera (2022)	2022	- London LEZ reduced average daily PM_{10} levels by 18 %, insignificant effect on NO_2 - London ULEZ reduced average daily PM_{10} by 27 %, NO_2 by 12.4 %
Has the ultra low emission zone in London improved air quality?	Ma et al. (2021)	2021	London ULEZ caused average reduction of less than 3 % for NO_2 levels, insignificant effect on O_3 and $PM_{2.5}$ concentrations
Did the London congestion charge reduce pollution?	Green et al. (2020)	2020	London Congestion Charge Zone caused substantial reductions in PM_{10} levels (around 20 %), NO (around 20 %), and CO (around 8 %), whilst causing increases in NO_2 levels
Five years of London's low emission zone: Effects on vehicle fleet composition and air quality	Ellison et al. (2013)	2013	London LEZ caused mean concentration of PM_{10} to decrease by average of 13 % per year, no significant difference in NO_x concentrations
Pollution and congestion in urban areas: The effects of low emission zones	Bernado et al. (2021)	2021	Across European LEZs, impact on pollutant levels were marginal, with ranges from -2.5 % to -8.5 %
Health effects of Low Emission Zones: Evidence from German hospitals	Pestel and Wozny (2021)	2021	German LEZs reduced the annual mean concentration of PM_{10} by 5.8 %, NO_2 by 5.3 %
Low emission zones and population health	Margaryan (2020)	2020	German LEZs reduce monthly PM_{10} concentrations by 2 % to 3 %, insignificant reductions in NO_2 concentrations
The effect of low emission zones on air pollution and infant health	Gehrsitz	2016	- Most restrictive German LEZs reduce $PM_{2.5}$ pollution by 6 % to 8 %, PM_{10} by 3 % to 4 % - Average German LEZ reduce $PM_{2.5}$ pollution by 4 %, PM_{10} by 2.5 %
Keep Your Clunker in the Suburb: Low-emission Zones and Adoption of Green Vehicles	Wolff (2014)	2014	German LEZs decrease PM_{10} concentrations by 9 %

observed minor improvements in air quality in London. Studying the effect of London's 2008 LEZ after five years, Ellison et al. (2013) found that there were no discernible in NO_x concentrations, with mean concentrations decreasing by an average of just 0.5 % to 1.5 % per year. However, the five-year analysis of PM_{10} trends yielded much larger reductions, with mean concentrations of PM_{10} decreasing by an average of 13 % per year. This study also examined vehicle turnover rate, and found an additional 20 % drop in pre-Euro 3 rigid vehicles in London due to the LEZ policy. The hypotheses that LEZs do trigger significant switches towards cleaner vehicles is backed in this report, as it finds that the vehicle turnover rate for affected vehicle classes in London returned to the national average after initial substantial increases. Following analysis of the spatial substitution in vehicle's emission categories, Wolff (2014) also concludes that LEZs do generate significant increases in the adoption of cleaner vehicles; the rate of drivers adopting green

commercially used vehicles increases by 88 % (and privately used green vehicles by 5 %) the closer they live to an LEZ.

There are also an extensive body of research into the impact LEZs have had on improving the health of residents. Analysing the effect of London's policies, [Beshir and Fichera \(2022\)](#) found significant reductions in limiting health problems. It found that the LEZ reduced overall limiting health problems by 7 %, with an especially significant impact on Chronic Obstructive Pulmonary Disease (14.5 %). Utilising the same approach to the London ULEZ, it found that limiting health problems were reduced even more (by 22.5 %). [Pestel and Wozny \(2021\)](#) found that German LEZs triggered 'small but statistically significant' health benefits; driven by reductions in circulatory and chronic lower respiratory diseases. There are a wide variety of health outcomes associated with air pollution, including the effect on infant health. [Gehrsitz \(2017\)](#) found that the German cities with the strictest LEZs were the most effective in reducing air pollution, with $PM_{2.5}$ decreasing by about 6 % to 8 % and PM_{10} decreasing by about 3 % to 4 %. Studying the impact of LEZs curbing pollutant levels, it concluded that neither average birthweight nor the prevalence of low-weight births were significantly affected by LEZs, although there were minor reductions in the incidence of stillbirths. [Margaryan \(2020\)](#) examines the links between pollution and respiratory and cardiovascular health, finding that that the health impacts of LEZs are particularly significant for people over 65. Using difference-in-differences design on outpatient and inpatient health care statistics, this report found that the number of people with cardiovascular disease reduced by 2–3 % following the introduction of LEZs, whilst there were insignificant reductions in respiratory-related hospitalisations.

Overall, evidence from existing literature suggests that LEZs and similar policies do trigger improvements in public health. Another critical aspect of LEZ policy is the location of the zone boundaries. [Pestel and Wozny \(2021\)](#) finds that there are limited spillovers of health benefits to hospitals outside of zone boundaries, suggesting that applying restrictions to whole city-regions (e.g. the London ULEZ following its expansions) may be more beneficial than only applying them to inner city regions. Research from existing literature also highlights the extensive economic impact caused by savings in the healthcare sector following LEZs reducing air pollution. For example, [Wolff \(2014\)](#), found that vehicle fleet turnover triggered by German LEZs resulted in health benefits of roughly of roughly \$2 billion USD. Data tables provided in the [Webber et al. \(2020\)](#) report, which models the long-term health benefits of the ULEZ in London, are utilised to answer RQ2. This report also models the economic impact of the ULEZ, calculating that it has a cost saving to NHS and social care system of £4.2 billion.

It should be noted that pollutant concentration levels, and long-term health benefits, also depend on how well low-carbon transport modes, such as walking and cycling, are promoted to the public ([Cieřla and Macioszek, 2022](#); [Macioszek et al., 2022](#)). With greater awareness and promotion of non-vehicular options, the potential for greater improvements in pollutant concentration levels, and thus also long-term health benefits, also exist.

This study contributes to the existing literature on Low Emission Zones by focusing on Manchester's Clean Air Policy, an area with limited prior research, and modeling its potential impact based on data from other cities. The comparison between the effects of three separate Low Emission Zone policies in London is also a research contribution, with the results indicating what types of policy are the most effective. Additionally, this study includes the analysis of $PM_{2.5}$, a more harmful pollutant than PM_{10} , contributing towards offering a more nuanced understanding of the health impacts of LEZs.

2. Materials and methods

2.1. Research Question 1

RQ1 (Does the implementation of Low Emission Zones in urban areas

significantly reduce air pollution?) assesses the trends in NO_2 and $PM_{2.5}$. Theil-Sen analysis is conducted across sites in London and Manchester.

Data for answering the first research question is taken from the Automatic Urban and Rural Network (AURN), which is the UK's largest automatic air quality monitoring network. There are currently 175 active sites ([DEFRA, 2024](#)) across the UK, including 16 sites across Greater London and 7 across Greater Manchester. Reasons for choosing the AURN sites include that the range of monitoring sites are sufficiently diverse and includes sites in Manchester (unlike the London Air Quality Network, run by the Environmental Research Group). The active AURN sites have live hourly pollutant readings, allowing most recent ULEZ to be analysed.

To assess the impact of the London LEZ (2008), London ULEZ (2019), and the first ULEZ Expansion (2021), different data frames were created for each site. This study does not include primary analysis for the T-charge introduced in 2017 due to its less stringent restrictions and proximity to the ULEZ. Likewise, the second ULEZ expansion introduced in August 2023 is excluded from the scope, due to not enough time passing since the policy was enforced. [Fig. 1](#) below shows the dates used to conduct analysis for Period 1 (LEZ), Period 2 (ULEZ), and Period 3 (ULEZ Expansion). Note that data frames for before and after phases (e.g. Phase 2a) are also included. For the Manchester sites, Period 4 is created to analyse trends up to the cancelled CAZ in May 2022. Period 5 is created to compare the results from Period 4, and includes data in the run up to the London ULEZ.

[Table 2](#) presents the selected sites, which vary in their classification. Traffic stations tend to record higher emissions due to close proximity to busy roads; these stations by definition are usually at least 25 m from the edge of major junctions and no more than 10 m from the kerbside ([DEFRA, 2024](#)). By contrast, background (and industrial) stations are composed of the integrated contribution of many sources in a several kilometre area ([DEFRA, 2024](#)). A total of six AURN sites are analysed for London and four for Manchester, with the option of using more restricted by the availability of NO_2 and $PM_{2.5}$ pollutant data over the Time Periods. [Figs. 2 and 3](#) illustrate the different zones in London and Manchester, as well as the location of the sites.

The openair package in R ([Carslaw and Ropkins, 2012](#)) was used to analyse the trends in NO_2 and $PM_{2.5}$ concentrations across the 10 sites. It has been developed over time to have advanced capabilities of manipulating air pollution data and producing sophisticated quantitative trend analysis outcomes. Theil-Sen analysis is to detect trends. This method estimates the median of all pairwise slopes between data points, and is useful for data that may have outliers or is not normally distributed. Advantages of using Theil-Sen analysis include that bootstrap sampling is used, which increases the robustness of the parameters used. When used on linear trends, Theil-Sen analysis tends to yield accurate confidence intervals even with non-normal data and heteroscedasticity ([Carslaw and Ropkins, 2012](#)). The results are also typically resistant to outliers, and the data can be 'deseasoned' (ie. ignoring seasonal variations that can potentially skew results) in the openair package. However, statistically significant trends may not be detected for short datasets; preventing several sites from being used. Nevertheless, Theil-Sen is widely used in environmental science because environmental data often exhibit non-normality, outliers, and heteroscedasticity. When analyzing trends in air quality, precipitation, temperature, or other environmental variables, Theil-Sen offers a reliable method to detect trends and changes over time. Plots using Theil-Sen analysis are shown in [Section 4](#), including plots produced using the openair 'slope.percent' function, which is useful for comparing sites that have very different concentration levels.

To further understand the impact that LEZs have on reducing particulate emissions, further analysis in Openair was completed to understand the behaviour of the two pollutants. For each site and Period, the dispersal of pollutants and ratio of the concentrations of $NO_2/PM_{2.5}$ with respect to each other were measured using dilution line plots and pollutant ratios. The RQ1 model introduces control variables for these

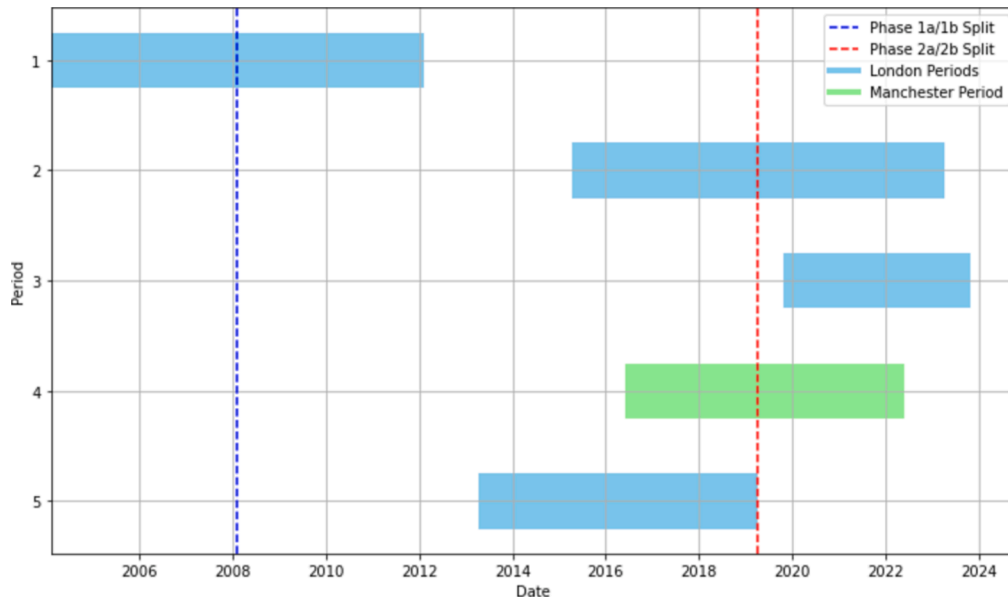


Fig. 1. Timeline showing the different time periods used in the analysis.

Table 2
AURN sites.

City	Site Name	Reference on Map (colour)	Classification
London	Marylebone Road	Red	Urban Traffic
London	Camden Kerbside	Blue	Urban Traffic
London	Bloomsbury	Black	Urban Background
London	Harlington	Yellow	Urban Industrial
London	Eltham	Purple	Suburban Background
London	Bexley	Orange	Suburban Background
Manchester	Bury Whitefield Road	Red	Urban Traffic
Manchester	Manchester Piccadilly	Black	Urban Background
Manchester	Salford Eccles	Purple	Urban Background
Manchester	Sharston	Blue	Suburban Industrial

measures, as well as introducing control parameters based on further robustness checks following the creation of Generalised Additive Models (GAMs). For each site and Period, the GAMs produce indications of the significance and fit of the data, by calculating R-squared values, parametric coefficient estimates, generalised cross validation values, and scale estimation values. GAMs are a preferred choice for trend analysis due to their flexibility in modeling non-linear relationships, ability to handle multiple predictors and seasonal patterns, robustness to outliers, and ease of interpretation. They are especially useful for analysing real-world datasets with complex, irregular patterns.

2.2. Research Question 2

RQ2 (Does the implementation of Low Emission Zones in urban areas significantly improve population health?) assesses the long-term effectiveness of LEZs on various health outcomes. Regression analysis is conducted across boroughs in London.

As previously discussed, data for hospitalisation and disease incidences are normally used to quantify the health impacts of LEZs. However, access to publicly available data in the UK filtered by diseases is restricted. The Webber et al. (2020) report, which uses micro-simulation modelling to attribute new disease cases to three different

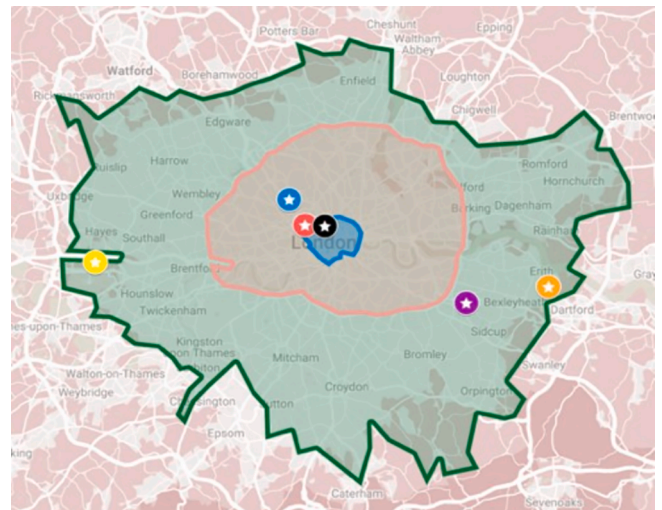


Fig. 2. Map of Greater London showing original LEZ/ULEZ boundary (blue), ULEZ expansion boundaries (red/green). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

scenarios (baseline, ULEZ, London Environment Strategy) from 2016 to 2050 was identified as the most appropriate source of data. Others were considered, including the quarterly Labour Force Survey and NHS online, however such sources featured heavy limitations (e.g. only overall summaries were made available to the public). The Webber et al. (2020) model uses NO_2 and $PM_{2.5}$ concentration data from the London Atmospheric Emission Inventory (LAEI) in 2013, and utilises geography-specific population and disease statistics from the Office of National Statistics (ONS) to produce results for long-term health indicators broken down into boroughs. Disease incidence data from the Webber et al. (2020) report was analysed for four boroughs (Bexley, Camden, Hillingdon, Kensington). These boroughs were chosen due to their location (covering all AURN sites used in RQ1). The nine health associations that have been linked to over-exposure to NO_2 and $PM_{2.5}$ emissions are shown in Table 3. Six diseases are used to analyse trends in diseases attributable to NO_2 , and seven for $PM_{2.5}$.

The Webber et al. (2020) modelling is highly detailed, utilising

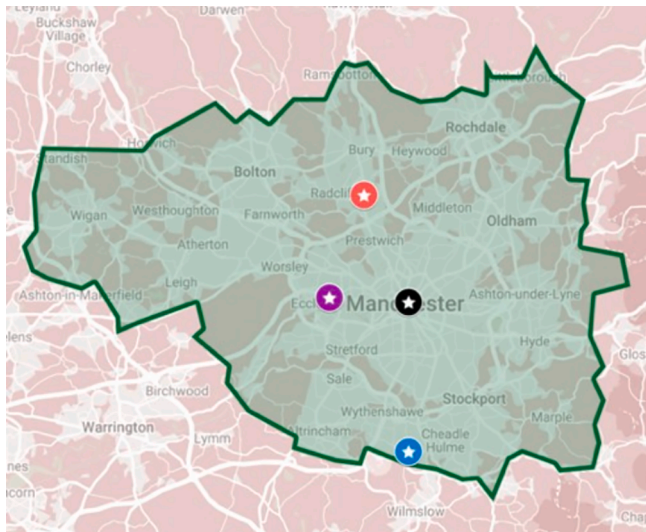


Fig. 3. Map of Greater Manchester showing CAZ boundary (green). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

population and air pollution data from different online databases, and new disease cases (incidence) and exposure–response coefficients from various literature reviews. For each pollutant and disease, data for incidence from 2016 to 2050 was extracted from the individual borough files available from the Webber et al. (2020) report. These incidence cases were then analysed using Python to quantify the overall difference between each the baseline (no change) and ULEZ scenario. An example plot for the Camden borough is shown in Fig. 4, demonstrating how the restrictions of the London ULEZ (and proposed LES) are modelled to reduce the incidence of asthma in children. The LES scenario (which includes extra restrictions on top of the ULEZ policy) is included in Fig. 4 but is not included in the scope of this report.

For each disease, integration performed by Python analysis was completed to calculate the accumulation of disease incidence for each scenario. Results for the percentage change of the ULEZ scenario compared to the baseline scenario were then computed for each disease. To ensure the integration performed by Python did not just calculate the ‘area difference’ between different cases, controls were included in the code to ensure any events where the two trend lines in question crossed over did not skew the results. Two control variables are introduced in RQ2: relative association score (s_1) and relative severity score (s_2). The relative severity score indicates the relative risk each disease carries, and is based on secondary mortality data from the Department of Health and Social Care public health profile tables (PHE, 2024). The relative association score indicates the relative risk long-term exposure to $NO_2/PM_{2.5}$ has on each respective disease, and is based on coefficients from the Webber et al. (2020) public health profile tables (PHE, 2024). The relative association score indicates the relative risk long-term report.

Table 3

Diseases associated with exposure to NO_2 and $PM_{2.5}$.

Disease	Associated with NO_2 ? (Y/N)	Associated with $PM_{2.5}$? (Y/N)	s_1 (NO_2)	s_2 (NO_2)	s_1 ($PM_{2.5}$)	s_2 ($PM_{2.5}$)
Asthma (children)	Y	Y	−0.60	+0.50	−0.87	0.72
Asthma (adults)	Y	N	−0.20	+0.45	—	—
Diabetes	Y	Y	−0.60	+0.46	+0.95	+0.69
Low Birth Weight	Y	Y	−0.20	+0.46	+0.44	+0.68
Lung Cancer	Y	Y	+0.60	−0.36	+1.0	−0.30
Dementia	Y	N	+1.0	−1.51	—	—
Coronary Heart Disease	N	Y	—	—	−0.53	−1.28
Stroke	N	Y	—	—	+0.81	−0.33
Chronic Obstructive Pulmonary Disease	N	Y	—	—	−0.92	−0.17

The respective control coefficients are included in Table 3 below.

3. Theory and calculation

For RQ1, Eq. (1) is the regression model used to calculate the overall change in pollutant levels observed in each site. The model incorporates the Theil-Sen trend differences observed using the openair package (TS_1 , TS_2 , TS_3), as well as the adjusting the results for the six control parameters (r_1 , r_2 , z_{1-4}) using the regression parameters.

$$Y_{NO_2/PM_{2.5}} = \beta_1 TS_1 + \beta_2 (TS_3 - TS_2) + \beta_3 r_1 + \beta_4 r_2 + \beta_5 z_1 + \beta_6 z_2 + \beta_7 z_3 + \beta_8 z_4 \quad (1)$$

To find accurate observations that indicate the pre-post impact for each LEZ, the overall Theil-Sen trend difference (TS_1) is used, as well as the difference between the post-implementation trends (TS_3 , e.g. from Period 2b) and pre-implementation trends (TS_2 , e.g. from Period 2a). The regression parameters β_1 and β_2 are used to give heavier weighting to the overall Theil-Sen difference (TS_1), due to the effects caused by LEZs (e.g. upgrading vehicles) being observed before they are implemented. The four control parameters obtained from creating the GAMs are used to give weighting to observations that have more robust trends. For example, a higher z_1 (R^2 value) parameter indicates a better fit of the regression trend to the observed data for the pollutant in question. The control parameters z_{1-4} are mediated by the regression parameters β_{5-8} to minimise their significance in the overall outcome (ie. the main results are from the Theil-Sen trends). The control regression parameters are also varied according to the direction of the trend to avoid examples where a positive Theil-Sen trend has its results skewed in the wrong direction.

$$Y_{NO_2} = \sum \left(\frac{Y_{NO_2}}{x} \right) \quad (2)$$



Fig. 4. Graph showing the modelled incidence of childhood asthma in Camden for three scenarios.

$$Y_{PM2.5} = \sum \left(\frac{Y_{PM2.5}}{x} \right) \quad (3)$$

Eqs. (2) (NO_2) and (3) ($PM_{2.5}$) are used to find the overall percentage change that each of the three London Periods in question have for the respective pollutants. An average value (Y_{NO2} or $Y_{PM2.5}$) is found for the Theil-Sen trend per year (in $\mu g/m^3$). For each period and for each pollutant, a final value for the approximate percentage reduction observed is obtained by dividing (Y_{NO2} or $Y_{PM2.5}$) by the mean annual pollutant concentrations in London for the respective period. These mean annual pollutant concentrations are also obtained using openair. Using the same method, results for the mean annual pollutant levels for Period 4 (in Manchester) and Period 5 (in London) are collected. These observations are critical to determining the effectiveness of London's strict LEZ policies and producing likely outcomes that would be observed if similar policies were adopted in other UK cities.

3.1. Step-by-step description for RQ1

Data were collected from the Automatic Urban and Rural Network (AURN) for pollutant concentrations at sites in London and Manchester. Theil-Sen trend analysis was applied using the openair package in R to calculate trends in NO_2 and $PM_{2.5}$ concentrations before and after LEZ implementation. A regression model was constructed, incorporating Theil-Sen trend differences for overall, pre-implementation, and post-implementation trends (see Equation (1)). Control parameters from Generalised Additive Models (GAMs) are summarised in Table 4 below. The final model assigned greater weighting to Theil-Sen differences, adjusting the regression parameters to increase the influence of the critical Theil-Sen trend differences. Final results compare the trends across different LEZ implementation periods, highlighting variations in pollutants across the sites.

3.2. Model for RQ2

A similar approach is taken for the construction of the model used to answer RQ2. Using the long-term trends of nine diseases, a final percentage estimate of the overall impact of NO_2 and $PM_{2.5}$ on public health is computed. Equation (4), which is used for each disease, is the regression model used to quantify the difference in health effects observed at each site. Using the data from the Webber et al. (2020) report and running it through Python analysis, the total expected percentage reduction in disease incidence (I) is the main parameter. Two control parameters are introduced (s_1 , s_2), indicating the relative association score and relative severity score for each disease.

$$y_{NO2/PM25} = I + \beta_1 s_1 + \beta_2 s_2 \quad (4)$$

The control parameters (s_1 , s_2) are adjusted by the regression parameters (β_1 , β_2). From the literature review, it was found that there are greater overall differences between the relative health risk of their

association to the two pollutants. Therefore, in the model (Eq. (4)) the relative severity score (s_2), is given a disease than their heavier weighting by the regression parameters than the relative association score (s_1). The models are used to find the final overall percentage change in health effects due to the introduction of the ULEZ, with respect to NO_2 (Eq. (2)) and $PM_{2.5}$ (Eq. (3)).

An average value (Y_{NO2} or $Y_{PM2.5}$) is found by combining the health effects of each individual disease (y_{NO2} or $y_{PM2.5}$) and dividing by the number of diseases measured for the pollutant in question. Using Openair trends for Period 4 and Period 5, the feasibility of comparing the impact of the ULEZ in London to the impact the CAZ would have had in Manchester is assessed.

3.3. Step-by-step description for RQ2

Disease incidence data related to NO_2 and $PM_{2.5}$ exposure was collected from borough-specific files provided in the Webber et al. (2020) report. A regression model was used to compute percentage reductions in disease incidence for each pollutant and disease (see Eq. (4)). Two control parameters were introduced: the relative association score, which reflects the link between pollutant exposure and disease, and the relative severity score, indicating the health risks of each disease. The control parameters were weighted, with the severity score receiving greater emphasis due to the stronger health risks linked to long-term pollutant exposure. The final percentage change in health outcomes was calculated for both NO_2 and $PM_{2.5}$ summarising the combined health effects across all measured diseases.

4. Results

The results of the model built to answer RQ1 are shown in Figs. 5 and 6. Period 2 (around the introduction of the ULEZ zone in 2019) has the most extensive air pollution reductions. NO_2 concentrations reduced by around $3.46 \mu g/m^3$ per year around the introduction of the ULEZ, reflecting a 9.78 % change when compared with the pollutant levels at the time. Likewise, the biggest $PM_{2.5}$ reduction was also observed around the 2019 ULEZ introduction; a rate of around $-1.14 \mu g/m^3$ per year was measured across the sites, reflecting a 10.28 % reduction (NB. this high percentage change is due to the magnitude of $PM_{2.5}$ (in $\mu g/m^3$) being consistently lower than the respective NO_2 magnitudes). The ULEZ expansion to the London Outer Ring Road in late 2021 also saw reductions in pollutant concentrations, with annual observations of -4.66% and -3.01% detected in NO_2 and $PM_{2.5}$ levels respectively. The availability of $PM_{2.5}$ data from the Automatic Urban and Rural Network sites in London around the introduction of the LEZ in 2008 demands caution when discussing the results from the model; an increase of around 0.90 % was observed (equivalent to an increase of around $0.16 \mu g/m^3$ per year). All sites but one showed marginal increases in $PM_{2.5}$ concentrations, although not all sites were included in the model for the Period 1 LEZ analysis due to a lack of pre-2008 $PM_{2.5}$ data. An observation of a 2.14 % reduction in NO_2 levels was also found around the LEZ rollout, with this rate being less than in Periods 2 and 3.

Values for Y_{NO2} and $Y_{PM2.5}$ were calculated for Period 4 across Manchester. Leading up to May 2022, when the Manchester Clean Air Zone was cancelled, there was an annual reduction of around 8.07 % in NO_2 concentrations and 4.78 % in $PM_{2.5}$ concentrations. The reductions of pollutant concentrations during this time could be driven by natural vehicle replacement, and the potential effects of residents and businesses replacing non-compliant vehicles in the expectation that the CAZ would be introduced. Furthermore, the effects of Covid-19 on reducing travel may have also been a contributing factor to reduced vehicle emissions. As seen in the Theil-Sen trends for the Manchester Piccadilly sites (Figs. 7 and 8), NO_2 and $PM_{2.5}$ concentrations appear to dip around the start of 2020.

This theory is supported by research from the Manchester Metropolitan University, which was studying the number of sites that

Table 4
Control parameters for RQ1:

R^2 value	Indicates how well the trend fits the observed data, providing a measure of goodness-of-fit. Higher R^2 values signify a stronger relationship between the predicted values and actual pollutant data, making the trend more reliable.
Generalised Cross Validation Score (GCV)	Evaluates model performance and prevents overfitting. Higher values indicate a higher likelihood of overfitting.
Scale Estimation Values	Assess variability in the data, with higher values indicating data inconsistency.
Parametric Coefficient Estimates	Indicates the significance of individual predictors within the model, allowing for the adjustment of trends depending on the strength of each contributing factor.

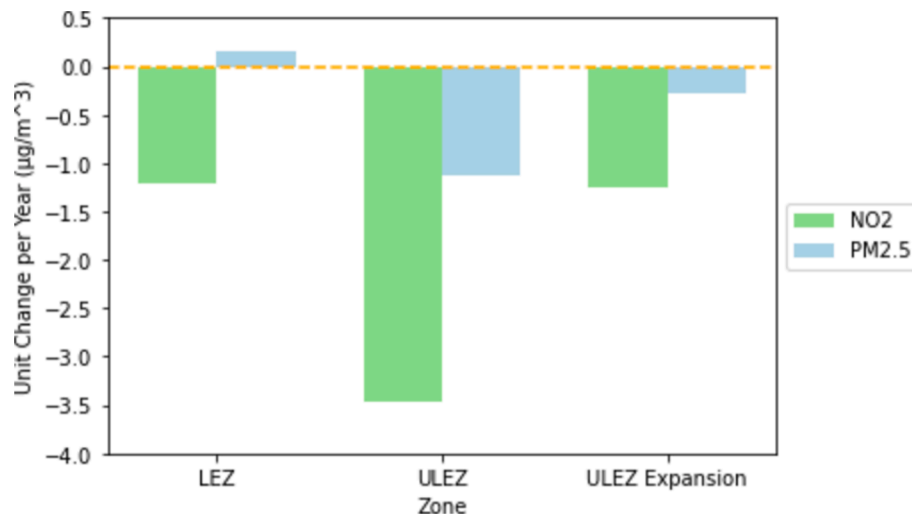


Fig. 5. Bar chart showing the overall pollutant unit changes per year for each London LEZ.

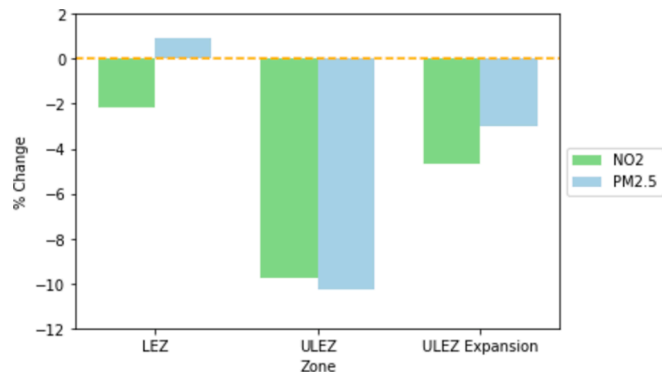


Fig. 6. Bar chart showing the overall pollutant percentage changes per year for each London LEZ.

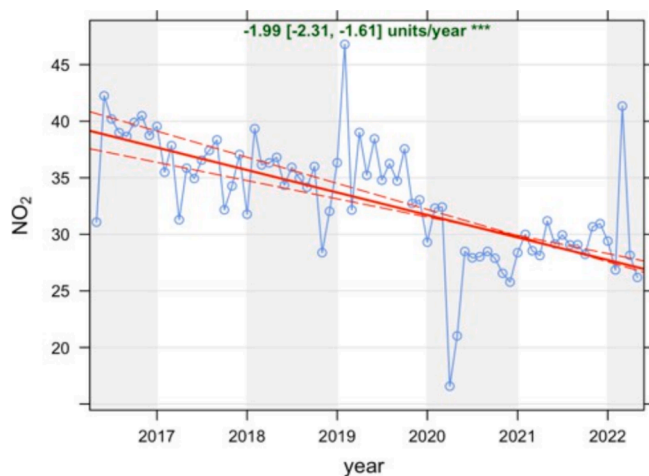


Fig. 7. Graph showing the Theil-Sen trends for the Manchester Piccadilly site in Period 4 for NO_2 .

exceeded the annual threshold for NO_2 around the 2020 lockdown. Using a total of nine sites, they found that number of sites exceeding the annual threshold reduced from six to four when lockdown was implemented (Megson, 2020).

The results for RQ2 are presented in Fig. 9 (NO_2) and Fig. 10 ($PM_{2.5}$),

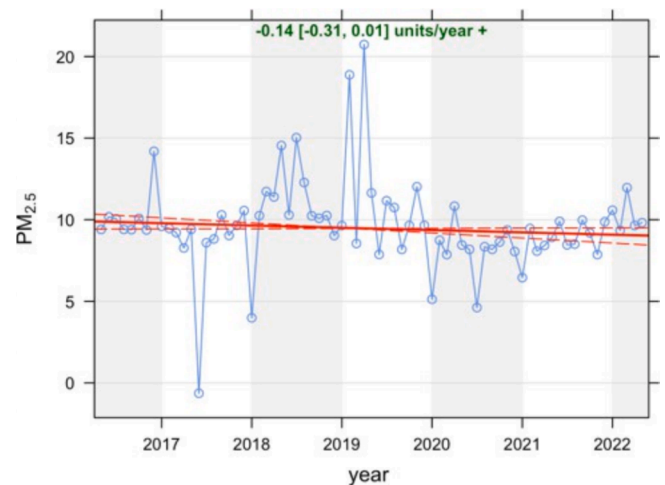


Fig. 8. Graph showing the Theil-Sen trends for the Manchester Piccadilly site in Period 4 for $PM_{2.5}$.

which show the expected percentage reduction in health outcomes in the period from 2016 to 2050 caused by the ULEZ scenario. The boroughs of Kensington and Camden, which are in central London (on or near the sites of Marylebone Road, Bloomsbury and Camden), are observed to have the greatest cuts in the incidence of the various diseases than the boroughs located in the outer London areas. The differences in disease cases between the sites in outer London and inner London are magnified for NO_2 when compared with the $PM_{2.5}$ effects. In Kensington and Camden, the ULEZ policies are found to create reductions of around 5.4 % and 5.1 % due to reductions in NO_2 , whereas in Hillingdon and Bexley these figures drop to 2.7 % and 2.3 %. In contrast, for the change in Diabetes incidence attributable to $PM_{2.5}$ decreases across the four boroughs; the reductions are lower and more consistent (at 1.7 %, 1.8 %, 1.3 % and 1.2 % respectively). Overall, this suggests that in the most densely populated areas of London, the ULEZ is more effective in improving health by means of cutting NO_2 emissions. In contrast the improvements by means of respectively). Overall, this suggests that in the most densely populated areas of London, the ULEZ is more cutting $PM_{2.5}$ emissions are only slightly larger in inner London.

Using the model presented earlier, it was possible to produce estimates of Y_{NO_2} and $Y_{PM_{2.5}}$ due to the harmful health effects for both pollutants. For harmful disease cases attributable to NO_2 , the ULEZ zone is introduction of the ULEZ zone (and its expansions). The results show

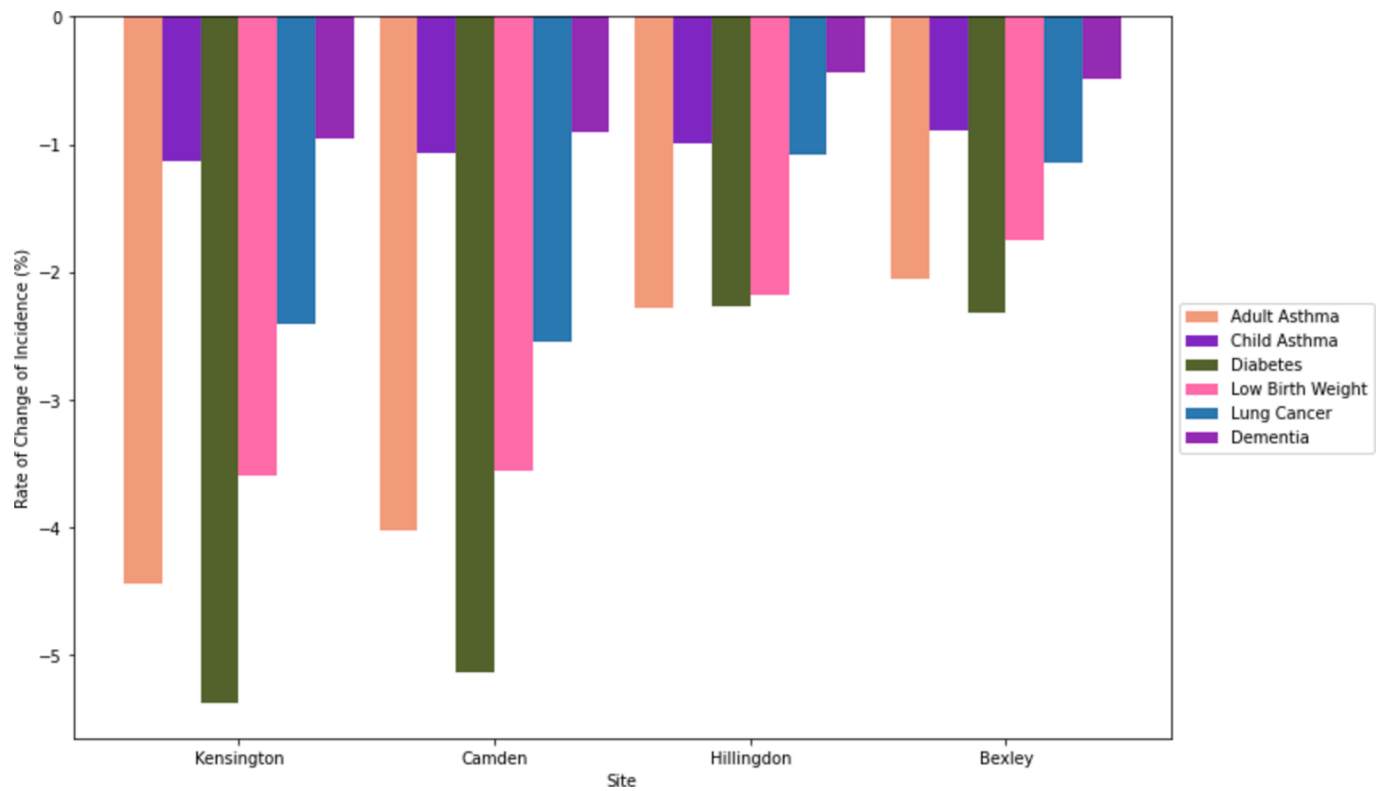


Fig. 9. Bar chart showing the rate of change of incidence of various diseases attributable to NO₂ reductions due to the ULEZ zone.

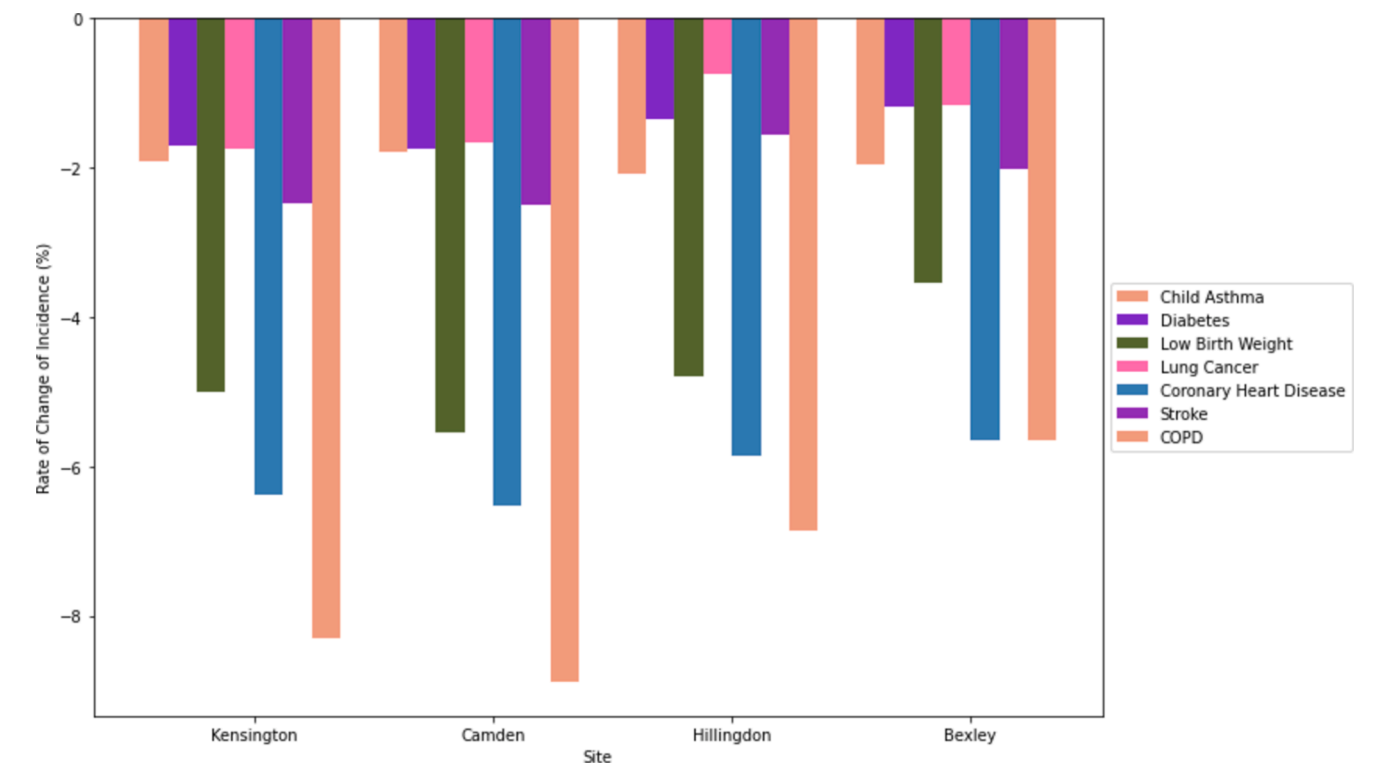


Fig. 10. Bar chart showing the rate of change of incidence of various diseases attributable to PM_{2.5} reductions due to the ULEZ zone.

that the ULEZ was effective in reducing found to reduce the overall number by around 2.24 %. For PM_{2.5}, this figure is higher at 3.56 %. These figures likely show that the ULEZ has a statistically significant effect on long-term public health.

To compare the air pollution and respective health impacts from the London ULEZ zone to Greater Manchester, several assumptions were made. Results using openair for Period 4 (leading up to the planned Manchester CAZ) show that annual mean concentrations for NO₂ were

25.48 $\mu\text{g}/\text{m}^3$ and for $\text{PM}_{2.5}$ were 9.66 $\mu\text{g}/\text{m}^3$. In Period 5 (leading up to the introduction of the London ULEZ in 2019), the respective values for NO_2 were 44.90 $\mu\text{g}/\text{m}^3$ and for $\text{PM}_{2.5}$ were 12.90 $\mu\text{g}/\text{m}^3$. Whilst the $\text{PM}_{2.5}$ pollution levels are relatively consistent between London and Manchester, there is a significant difference between the annual mean NO_2 levels, with London being above the UK legal limit of 40 $\mu\text{g}/\text{m}^3$ and Manchester being significantly below (at 25.48 $\mu\text{g}/\text{m}^3$). However, these results may be skewed due to the availability of robust data for the various sites; NO_2 observations during this time at the London sites of Marylebone Road and Camden are unusually high, with Marylebone Road experiencing an extremely high average of 87.13 $\mu\text{g}/\text{m}^3$. Considering the likely effect Covid-19 had on causing air pollution to fall below normal levels for the Manchester sites only (NB. the run up to ULEZ was before the Covid-19 pandemic), we assume that London's 'before ULEZ' overall mean air pollution concentrations were similar to Manchester's 'before CAZ' air pollution levels. After analysing Theil-Sen trends using the slope percentage change function, the same assumption is made for the direction of the pollutant trends between the two cities.

It is assumed that the air pollution and health observations in London triggered by the introduction of the ULEZ zone in 2019 is a good indicator to the potential effects the introduction of the CAZ in 2022 would have likely had in Manchester. A key part of this assumption is that Manchester has never enforced a specific charging policy aimed at restricting polluting vehicles, whereas London had already introduced the LEZ prior to the ULEZ. Therefore, despite the ULEZ restrictions being much tougher than the CAZ (e.g. impacting cars as well as HGVs and vans), the overall changes in pollution concentrations may be similar due to the similar magnitude of change.

Overall, it is observed that the enforcement of the Manchester CAZ would have caused NO_2 and $\text{PM}_{2.5}$ rates to likely experience annual reductions of around 10 % for both NO_2 and $\text{PM}_{2.5}$ concentrations if it had introduced fall well below the UK legal limits. Like the effect the introduction of the London ULEZ had, Manchester would fleet turnover. The likely impact of the CAZ would be especially beneficial to reducing $\text{PM}_{2.5}$ concentrations, the CAZ. Note that, like in London, this reduction is in the context of an overall downward trend due to natural which are currently regularly above the UK legal limits in Manchester (see Fig. 6b). Like in London, $\text{PM}_{2.5}$ levels would likely approach the limits set by the World Health Organization, which recommend that annual mean levels do not rise above 5 $\mu\text{g}/\text{m}^3$ (WHO, 2021). Due to the strong association of several diseases with over-exposure to $\text{PM}_{2.5}$, a curbing in $\text{PM}_{2.5}$ concentrations would also likely bring statistically significant improvements in long-term health. RQ2 results suggest that the introduction of new policy in Manchester could cause positive outcomes like those observed in London (a 2.24 % reduction in the incidence of diseases attributable to NO_2 and a 3.56 % reduction in the incidence of diseases attributable to $\text{PM}_{2.5}$).

5. Discussion

The results from the three London policies analysed using Theil-Sen trends indicate that Low Emission Zones are effective in reducing urban air pollution. The analysis shows that the ULEZ (and its subsequent expansion 2021) was the most effective in reducing NO_2 and $\text{PM}_{2.5}$ concentrations, which are now consistently below the UK legal guidelines in most sites. However, the LEZ introduced in 2008 was less effective in reducing pollution. Whilst small reductions were observed around the LEZ introduction for NO_2 , $\text{PM}_{2.5}$ concentrations slightly increased by around 0.90 %. This may be attributable to a lack of suitable $\text{PM}_{2.5}$ data around this time, as other studies show different outcomes. Ellison et al. (2013) found that overall pollution levels within the London LEZ dropped by around 2.46 % to 3.07 %. Another observation from the RQ1 modelling for London is that the inner London sites had the biggest pollution reductions. This is positive towards the case for LEZs, as the central areas of cities are more likely to have dangerous levels of air pollution. Furthermore, the expansion of the ULEZ to cover

most of Greater London in 2023 is likely to ensure residents in the outer London areas are not exposed to unsafe pollution levels.

When analysing trends in air pollution, it is important to consider the impact of abrasion emissions. The European Environmental Agency warns that 46 % of pollutants generated by private transportation are generated by the so-called abrasion emissions (EEA, 2023). These emissions, which are a component of Particulate Matter, are generated by 'clean' cars as well as cars that do not meet emission standards. This raises the question of how LEZ restrictions should be adapted once nearly all cars meet the current standards, particularly since electric cars are heavier than internal combustion cars (thus causing greater abrasion emissions when breaking and accelerating).

The RQ2 results suggest that the ULEZ policies are more effective in reducing the long-term health effects for $\text{PM}_{2.5}$ (3.56 %) than NO_2 (2.24 %). This impact is critical towards the case for LEZs, as it has been well established in recent decades that over-exposure to $\text{PM}_{2.5}$ has detrimental effects on health. Webber et al. (2020) found that as many as 16 % of new disease cases by 2050 would be attributable to $\text{PM}_{2.5}$ in London if the ULEZ policy was not enforced (compared to 9 % for NO_2). This is due to the over-exposure of $\text{PM}_{2.5}$ significantly increasing the risk of diseases such as asthma and lung cancer. In combination with the serious health consequences high NO_2 pollution carries, reductions of around 3 % in the incidence of associated diseases is highly consequential to the long-term and health of residents. When applied across large populations, significant positive social and economic effects are likely to be observed following improvements in air pollution. For example, whilst studying the costs of NO_2 and $\text{PM}_{2.5}$, Public Health England estimated in 2018 that the health and social care costs of these pollutants could reach as much as £5.3 billion by 2035 unless action is taken (PHE, 2018). Another argument towards the case for LEZs is that vulnerable people are at high risk of harmful diseases as a result of air pollution, including elderly people (e.g. dementia), pregnant women (e.g. low birth weight) and children (e.g. asthma).

The political discussion around LEZs and other policies is highly significant. Especially in recent years, this topic has become one of the main debates around regional policy (e.g. in the London mayoral election in May 2024). Whilst this study concludes that the long-term health impacts justify the implementation of such zones, critics will continue to argue that the benefits are spread disproportionately across society (e.g. there is a 'financial hit' for businesses and residents, through charges and/or the cost of replacing vehicles). Other important factors to consider around the implementation is the disputed spillover effects (e.g. pollution being 'exported' to areas just outside LEZ boundaries), and the cost of implementing such zones (e.g. the money spent on preparation for the Manchester CAZ). However, proponents for LEZs will also argue that the negative 'side effects' are outweighed by the positive ones. For example, higher provision of public transport is preferred in cities for long-term sustainability, and more active travel (e.g. cycling) is likely to improve both the physical and mental health of people. Furthermore, noise is the second largest environmental sensor in Western Europe (WHO, 2011), and LEZs can help combat this by reducing traffic.

There are several limitations to this study, several of which have prevented more robust statistical modelling and overall conclusions to be made. As mentioned in Section 3, the availability of suitable sites from the Automatic Urban and Rural Network (AURN) limited the number of Theil-Sen tests that could be completed. This was a particular limitation for $\text{PM}_{2.5}$ data in London around the introduction of the LEZ in 2008, with a high proportion of AURN sites having incomplete data. One possible reason for this is that the highly harmful health impacts of over-exposure to Fine Particulate Matter were not as widely acknowledged as other pollutants were at the time. There were several reasons why further data from other air pollution monitoring networks (particularly for $\text{PM}_{2.5}$ data around 2008) was not used. Other data sources had fewer availability of sites, and (like the AURN network) also featured gaps in the datasets when they were put through the openair package at the start

of the project. Additionally, time and resource limitations restricted the opportunity to conduct detailed analysis using alternative monitoring networks. For generating results for the datasets that had gaps in them, interpolation was carried out automatically in the openair package Theil-Sen trend analysis.

Overall, there was a roughly even split of traffic and background stations for both the sites in London and Manchester, which is ideal for the openair analysis to represent the pollutant ranges over areas (e.g. London Bloomsbury site) as well as the locations that are expected to have the most dangerous pollutant levels (e.g. London Marylebone Road site). A further limitation for RQ1 is that only two pollutants are measured, with trends for other important pollutants (e.g. O_3 , PM_{10}) discounted from the report. Limitations in RQ2 are centered around the unavailability of individual-level and aggregate hospitalisation data in London, with data from the literature review used instead. The data from the Webber et al. (2020) report meant that the results were 'back-calculated'; potentially introducing inconsistencies with the data quality across the two research questions. If suitable secondary disease and hospitalisation data were available in London, this could have improved the robustness and originality of the results. However, for several reasons (including restrictions around the access of medical data), this was not possible. However, the 'back-calculated' results using data from the Webber et al. (2020) report are reliable, due to the highly detailed modelling that was used to produce the data (see Section 2.2).

The study focuses on Greater Manchester and its scrapped Clean Air Zone due to the alternative route policy makers in Manchester have taken, and a lack of existing research in place. A limitation of using Greater Manchester is that not enough time has passed to assess the air pollution and health impacts since the CAZ was replaced with the investment-led Clean Air Plan. Therefore, results from London are used to quantify the likely impact the CAZ would have had, based on the assumptions made from NO_2 and $PM_{2.5}$ concentrations in Manchester. There are several factors that create uncertainties in the assumptions made in Section 4.3. For RQ1, factors such as the different population density and public transport provisions in Manchester when compared to London are likely to introduce errors in the assumptions made. For RQ2, similar errors are likely to be introduced by variations in population demographics and the quality of health services.

A sensitivity analysis, exploring the uncertainties in the assumptions made between the two cities, would provide more robust outcomes. For example, numerical modelling into the variations in the proportion of people who use public transport, demographic factors (such as population density comparisons), and a comprehensive investigation into the effect of Covid-19 on travel, would be required for this analysis. This study does not include this modelling due to time and resource limitations.

To conclude this study, there are several recommendations for future LEZ policy. These are derived from the research question results and the literature review. For Manchester, the introduction of a revised Clean Air Zone would likely yield positive outcomes. Due to Manchester being a smaller (and less dense) city than London, and that current air pollution levels are satisfactory with respect to the legal limits, introducing a policy as strict as the London ULEZ is not necessary. Instead, daily charges for the most highly polluting HGVs and vans (similar to the London LEZ introduced in 2008, and the original planned Manchester CAZ in 2022) would be justified due to the potential health benefits. Factors such as natural fleet turnover negate the demand to introduce charges for non-compliant cars and motorcycles. Like in the current Clean Air Plan, continuing with investment-led strategies (e.g. the new electric bus 'Bee Network') is also recommended.

For London, the ULEZ has been effective in mitigating air pollution, and it is not recommended that this policy is reversed. Instead, new schemes aimed at minimising the 'financial hit' the ULEZ has had on lower income residents could be introduced (e.g. a new incentive scheme for small businesses to replace older vehicles). Across cities in the UK, public support for LEZs could be boosted by introducing new

transparent policy relating to the revenue collected from charges and fines. For example, the revenue from each quarter could be used to fund higher frequencies of public transport routes, with this information effectively communicated to residents. The continued investment in public transport is also highly recommended for all cities, especially those where there are proposals to introduce LEZs but the quality of public transport options is poor (e.g. Greater Manchester). Public transport enhancements are also likely to reduce the modal share that cars have, potentially leading to positive long-term effects such as the reduction in abrasion emissions from cars (including LEZ-compliant cars).

6. Conclusions

After constructing regression models to answer the two research questions, this study has produced quantitative outcomes of the pollutant concentrations and long-term health benefits caused by introducing LEZs. Results have shown that LEZs do have a significant positive effect on both of these metrics. In particular, the London Ultra Low Emission Zone implemented in 2019 was successful in curbing Nitrogen Dioxide and Fine Particulate Matter concentrations by about 10 % annually. Results from the analysis into the other policies (the 2008 LEZ and 2021 ULEZ expansion) show marginal but still significant reductions in pollution levels. This includes the reductions of around 4.6 % and 3.0 % in the respective Nitrogen Dioxide and Fine Particulate Matter concentrations around the rollout of the ULEZ expansion in 2021. The results for Research Question 1 also highlight variations in between inner and outer London. The inner London areas showed the greatest reductions in pollution levels, particularly after the ULEZ was implemented. This suggests that the ULEZ was more effective in areas with higher baseline pollution, especially in densely populated regions like Kensington and Camden.

The reductions in pollution levels around the ULEZ implementation were shown to reduce the incidence of several associated limiting health outcomes by around 3 %, especially with diseases such as Diabetes and Chronic Obstructive Pulmonary Disorder. Discussion of the results and information gained from existing literature concludes that, despite the complex nature of the positive and negative impacts of LEZs, they are essential to curbing air pollution concentrations in urban areas that exceed safe limits. The results for Research Question 2 highlight the significance of implementing policy aimed at reducing harmful Fine Particulate Matter concentrations in urban areas, with reductions in disease incidence due to lower Particulate Matter levels found to be greater than those associated with Nitrogen Dioxide. Despite the post-COVID reductions in pollution concentrations, driven by factors including vehicle replacement and reduced travel, further measures in Greater Manchester would likely ensure all air pollution targets are satisfied, including the World Health Organization's targets for Fine Particulate Matter, which are not currently met.

Recommendations for future research into this area include detailed analysis of the short-term pollution and health impacts of UK air pollution policies that have been recently introduced, including the investment-led Manchester Clean Air Plan. High-level modelling of such policies that study the likely long-term effects would also be beneficial to shaping future air pollution policy. Other avenues for future research into areas that are outside the scope of this study include investigation into the impact of behavioural patterns on LEZ policy. For example, the 'latent demand' effect is where consumers lack the purchasing power to obtain desired goods or services. A future study into this effect and LEZ policy (e.g. the affordability and availability of electric cars and public transport options) could be useful for shaping future LEZ policy, with potential solutions (e.g. subsidies) recommended following the analysis of behavioural patterns.

CRediT authorship contribution statement

Stan Broster: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Kathryn Terzano:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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